

Modem

Analog vs digital

In analog modulation, the modulation is applied continuously in response to the analog information signal.

In digital modulation, an analog carrier signal is modulated by a discrete signal.

AM

630 Meter ranges from 472 KHz to 479 KHz.

80 Meter ranges from 3.5 MHz to 4.0 MHz.

the amplitude (signal strength) of the wave is varied in proportion to that of the message signal.

$$y(t) = x(t) * A_c \sin(2\pi f_c t)$$

FM

Frequency modulation (FM) is the encoding of information in a carrier wave by varying the **instantaneous frequency** of the wave.

$$y(t) = A_c * \sin[2\pi * (f_c + f_d * x(t)) * t]$$

assuming $x(t)$ is limited to the range ± 1 .

f_d is the frequency deviation: $f_d = A_m * K_f$

A_m is the amplitude of the modulating signal or baseband signal.

K_f is the sensitivity of the frequency modulator.

ASK

Amplitude-shift keying (ASK) is a form of amplitude modulation that represents **digital data as variations in the amplitude** of a carrier wave.

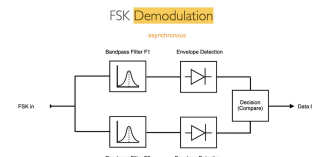
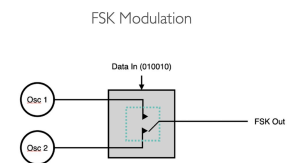
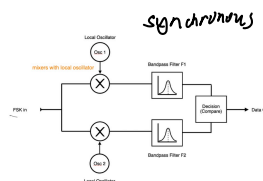
$$y(t) = [DC + m * f(t)] A_c \sin(2\pi f_c t)$$

$f(t)$: digital input

f_c is the carrier frequency, m is the modulation amplitude and A_c is carrier amplitude.

FSK

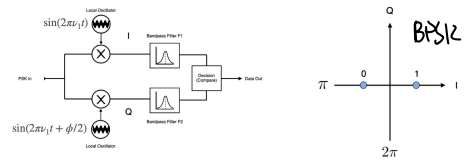
Frequency-shift keying (FSK) is a frequency modulation scheme in which **digital** information is encoded on a carrier signal by shifting the frequency of the carrier between several **discrete frequencies**.



PSK

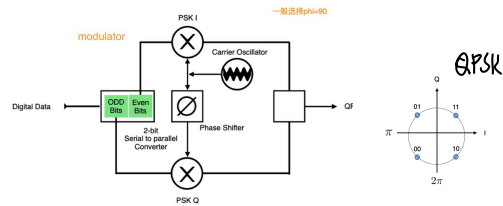
Phase-shift keying (PSK) is a **digital** modulation process which conveys data by changing (modulating) the **phase** of a constant frequency carrier wave.

Binary phase-shift keying (BPSK)

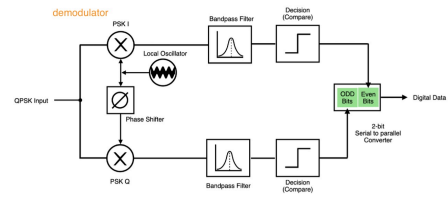
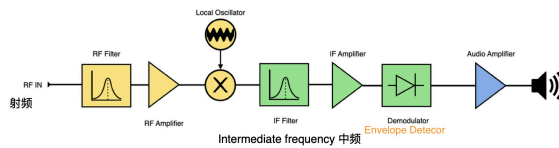


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Quadrature phase-shift keying (QPSK)



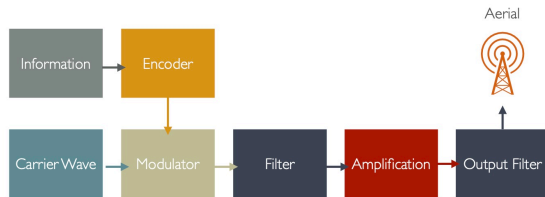
SuperHeterodyne receivers



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The Transmitter

The basic set of components in a transmitter are the following:



Transmitter